Layer	Truthful		Neutral		Deceptive	
	TTPD	LR	TTPD	LR	TTPD	LR
1	50.69 ± 1.10	49.50 ± 0.00	50.43 ± 0.65	49.50 ± 0.00	50.63 ± 1.00	49.50 ± 0.00
2	50.17 ± 0.35	49.56 ± 0.35	50.36 ± 0.66	49.48 ± 0.11	50.16 ± 0.36	49.48 ± 0.17
3	50.07 ± 0.08	49.64 ± 0.27	50.21 ± 0.19	49.79 ± 0.43	50.20 ± 0.36	49.76 ± 0.45
4	50.23 ± 0.31	50.62 ± 0.88	50.49 ± 0.50	50.74 ± 0.96	50.14 ± 0.24	50.11 ± 0.68
5	49.83 ± 0.63	51.20 ± 0.64	49.88 ± 0.40	51.08 ± 0.83	49.91 ± 0.56	50.95 ± 0.62
6	49.91 ± 0.55	50.92 ± 0.18	49.88 ± 0.47	50.93 ± 0.28	49.87 ± 0.24	50.84 ± 0.33
7	52.38 ± 1.31	52.02 ± 0.64	51.58 ± 1.48	52.63 ± 1.06	51.01 ± 1.70	52.52 ± 1.12
8	52.90 ± 1.26	52.13 ± 0.61	52.04 ± 0.86	52.94 ± 1.02	52.46 ± 0.75	52.12 ± 0.67
9	58.66 ± 1.96	53.09 ± 1.60	57.93 ± 1.99	56.53 ± 3.05	57.77 ± 2.41	53.88 ± 2.34
10	59.58 ± 4.57	61.37 ± 3.67	57.23 ± 4.96	62.20 ± 4.01	58.79 ± 3.42	62.05 ± 3.08
11	68.41 ± 0.74	57.16 ± 3.11	64.66 ± 2.49	59.57 ± 3.65	60.11 ± 3.96	57.07 ± 3.38
12	69.69 ± 0.63	63.11 ± 3.09	69.75 ± 0.83	62.33 ± 4.54	69.13 ± 0.73	58.76 ± 3.67
13	77.20 ± 0.23	65.37 ± 4.95	80.10 ± 0.24	62.92 ± 5.64	73.43 ± 0.83	62.25 ± 4.66
14	78.22 ± 0.23	75.70 ± 2.88	82.25 ± 0.53	71.66 ± 4.21	79.29 ± 0.30	70.44 ± 3.74
15	80.04 ± 0.47	75.97 ± 3.46	84.86 ± 0.47	75.49 ± 3.53	80.42 ± 0.13	70.07 ± 6.07
16	82.58 ± 0.60	77.15 ± 4.43	85.62 ± 0.14	74.91 ± 4.07	79.23 ± 0.11	71.33 ± 4.35
17	83.62 ± 0.41	76.97 ± 3.27	85.42 ± 0.15	74.03 ± 5.79	79.80 ± 0.29	69.46 ± 6.76
18	83.55 ± 0.35	74.53 ± 4.70	85.01 ± 0.30	74.43 ± 4.92	76.25 ± 0.35	67.11 ± 6.53
19	83.24 ± 0.22	73.62 ± 5.45	83.75 ± 0.34	73.71 ± 4.70	78.69 ± 0.18	67.65 ± 5.26
20	83.07 ± 0.41	74.90 ± 4.18	83.62 ± 0.22	71.85 ± 4.34	78.69 ± 0.48	66.16 ± 7.26
21	82.84 ± 0.27	69.61 ± 5.96	83.41 ± 0.36	76.20 ± 4.87	79.50 ± 0.34	66.87 ± 6.48
22	82.53 ± 0.35	71.29 ± 5.17	83.61 ± 0.21	73.31 ± 5.80	79.22 ± 0.23	66.44 ± 7.18
23	82.25 ± 0.29	73.93 ± 5.95	83.39 ± 0.24	72.20 ± 4.89	79.06 ± 0.37	67.68 ± 6.19
24	82.11 ± 0.37	71.52 ± 5.36	83.22 ± 0.24	72.48 ± 6.43	78.54 ± 0.33	68.94 ± 5.61
25	82.10 ± 0.29	73.07 ± 6.04	83.30 ± 0.33	72.60 ± 6.11	78.24 ± 0.19	67.35 ± 6.50
26	81.96 ± 0.29	71.62 ± 6.31	83.26 ± 0.19	73.14 ± 5.08	78.33 ± 0.26	70.58 ± 4.26
27	81.55 ± 0.28	71.34 ± 6.16	82.97 ± 0.23	73.90 ± 5.83	77.45 ± 0.56	69.06 ± 5.07
28	81.42 ± 0.33	73.38 ± 5.56	82.90 ± 0.24	75.66 ± 5.13	76.81 ± 0.30	68.20 ± 5.64
29	81.36 ± 0.18	75.02 ± 3.82	82.70 ± 0.25	73.47 ± 4.71	76.43 ± 0.38	68.37 ± 6.25
30	81.31 ± 0.22	70.84 ± 6.26	82.87 ± 0.34	72.73 ± 4.14	76.51 ± 0.60	69.63 ± 7.97
31	81.62 ± 0.23	71.57 ± 5.66	82.89 ± 0.33	71.96 ± 5.07	76.17 ± 0.71	63.63 ± 7.32
32	81.57 ± 0.33	64.56 ± 4.66	83.87 ± 0.19	72.36 ± 5.25	77.86 ± 1.23	68.55 ± 5.39

Table 3: Layer-wise probing accuracy for Llama3.1-8B-IT across truthful, neutral, and deceptive prompts using TTPD and LR.

Layer	Truthful		Neutral		Deceptive	
	TTPD	LR	TTPD	LR	TTPD	LR
1	50.47 ± 0.59	50.69 ± 0.51	51.03 ± 0.91	50.75 ± 0.61	50.54 ± 0.74	50.18 ± 0.47
2	51.13 ± 0.73	50.55 ± 0.68	51.45 ± 1.29	50.15 ± 0.66	50.73 ± 0.76	50.41 ± 0.80
3	50.80 ± 1.16	50.51 ± 0.70	51.68 ± 0.99	50.45 ± 0.65	51.07 ± 0.76	50.97 ± 0.77
4	51.24 ± 0.55	50.55 ± 0.69	51.22 ± 1.00	50.49 ± 0.49	50.98 ± 0.67	49.64 ± 0.61
5	51.04 ± 0.56	50.95 ± 0.51	50.84 ± 0.72	51.14 ± 0.30	51.40 ± 0.71	51.38 ± 0.60
6	51.50 ± 0.62	50.79 ± 0.39	50.75 ± 0.55	51.20 ± 0.58	51.35 ± 0.78	50.88 ± 0.35
7	48.31 ± 1.22	49.89 ± 0.82	48.09 ± 0.75	49.69 ± 1.17	48.17 ± 1.35	49.42 ± 0.90
8	49.17 ± 0.91	50.68 ± 1.09	49.75 ± 0.84	51.09 ± 0.63	48.43 ± 0.99	50.71 ± 0.67
9	52.31 ± 0.72	51.17 ± 0.71	52.45 ± 1.02	51.84 ± 1.05	52.99 ± 0.50	51.13 ± 0.62
10	52.74 ± 0.86	51.85 ± 0.58	51.93 ± 0.66	52.31 ± 1.02	53.31 ± 0.36	51.90 ± 0.54
11	52.98 ± 1.08	52.99 ± 0.78	51.92 ± 0.33	53.58 ± 1.63	53.37 ± 0.89	51.96 ± 0.67
12	51.72 ± 0.33	52.40 ± 0.87	51.74 ± 0.32	53.76 ± 0.87	53.00 ± 1.27	52.47 ± 0.65
13	52.63 ± 0.94	53.56 ± 0.99	52.11 ± 0.43	53.38 ± 1.52	52.95 ± 1.22	52.31 ± 0.78
14	53.74 ± 1.31	53.92 ± 1.09	54.00 ± 1.97	53.80 ± 1.49	53.28 ± 1.30	52.44 ± 0.52
15	54.80 ± 1.16	57.56 ± 2.03	54.32 ± 1.83	54.40 ± 1.81	54.11 ± 0.81	55.54 ± 1.31
16	57.54 ± 1.84	62.03 ± 2.06	56.46 ± 2.46	57.29 ± 3.36	57.61 ± 0.76	59.86 ± 1.87
17	60.84 ± 1.98	65.56 ± 1.83	58.78 ± 2.10	64.45 ± 2.30	60.17 ± 1.41	62.68 ± 1.38
18	66.32 ± 1.09	68.80 ± 1.28	64.44 ± 2.09	65.77 ± 2.68	64.41 ± 1.36	63.64 ± 2.78
19	75.17 ± 0.34	70.32 ± 5.99	77.33 ± 0.96	65.93 ± 6.63	69.10 ± 0.86	69.75 ± 4.53
20	78.82 ± 0.47	72.61 ± 6.49	82.78 ± 0.95	70.72 ± 4.39	76.47 ± 0.44	72.12 ± 6.80
21	82.52 ± 0.20	72.38 ± 6.30	84.72 ± 0.28	69.97 ± 6.46	84.16 ± 0.51	76.32 ± 4.30
22	83.85 ± 0.34	64.00 ± 6.21	76.57 ± 0.95	65.25 ± 6.95	83.50 ± 0.26	71.22 ± 5.84
23	83.63 ± 0.36	71.50 ± 7.20	77.85 ± 0.98	71.69 ± 5.75	82.76 ± 0.33	69.36 ± 7.19
24	85.27 ± 0.11	71.62 ± 7.54	83.21 ± 0.47	73.37 ± 6.95	87.13 ± 0.75	70.12 ± 7.54
25	85.40 ± 0.23	73.09 ± 6.16	84.82 ± 0.28	72.89 ± 6.10	84.49 ± 0.74	72.56 ± 7.26
26	86.06 ± 0.20	75.38 ± 6.02	86.14 ± 0.44	68.37 ± 6.16	85.15 ± 0.65	71.84 ± 7.07
27	85.77 ± 0.39	79.72 ± 4.74	84.89 ± 0.39	71.21 ± 5.94	86.70 ± 0.18	74.29 ± 5.96
28	85.82 ± 0.17	79.99 ± 5.13	85.01 ± 0.28	76.32 ± 6.01	86.39 ± 0.49	73.53 ± 7.26
29	85.90 ± 0.13	80.06 ± 5.60	85.26 ± 0.27	78.20 ± 5.66	83.64 ± 0.45	72.23 ± 7.11
30	85.58 ± 0.21	79.15 ± 5.15	84.56 ± 0.28	74.26 ± 5.50	79.60 ± 0.51	74.41 ± 5.91
31	85.37 ± 0.19	76.35 ± 6.26	84.64 ± 0.26	75.32 ± 6.12	75.75 ± 0.80	75.85 ± 7.14
32	85.68 ± 0.22	77.95 ± 6.29	84.98 ± 0.23	77.45 ± 4.96	74.61 ± 0.89	71.97 ± 7.13

Table 4: Layer-wise probing accuracy for Gemma2-9B-IT across truthful, neutral, and deceptive prompts using TTPD and LR.

C PCA Visualization Results

This section presents supplementary PCA visualizations to further illustrate the global geometry of the models' internal activations under different instructional prompts. As discussed in the main text, PCA is employed to project the high-dimensional hidden state activations (x_l) onto a 2D space, primarily for illustrative purposes. These visualizations help in assessing the separability of internal states corresponding to "True" and "False" outputs across Neutral, Truthful, and Deceptive conditions.

The figures below provide additional examples beyond those in Section 3.2, showcasing these dynamics for both LLaMA-3.1-8B-Instruct and Gemma-2-9B-Instruct on various datasets. Specifically, Figure 10 and Figure 11 demonstrate the PCA results on curated datasets (e.g., `sp_en_trans` for LLaMA and `cities` for Gemma). These typically show a clearer separation between True/False clusters as observed in Figure 4 for the `cities` dataset with LLaMA. In contrast, Figure 5 (which may correspond to Figure 5 in the main text showing `common_claim` and `counterfact` for LLaMA) and Figure 12 (showing similar complex datasets for Gemma, as in Figure 12) illustrate the challenges PCA faces with more complex, uncurated datasets where the True/False clusters often appear entangled due to feature superposition. These appendix figures offer a broader visual substantiation of how the geometric separability of truth-related representations can vary significantly with dataset complexity and model type.

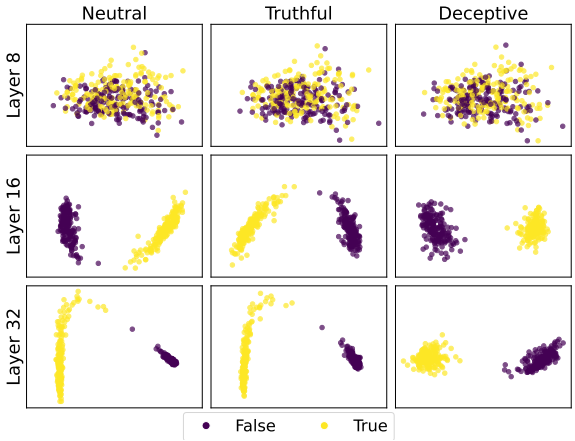

Figure 10: Layer-wise PCA visualization for LLaMA-3.1-8B-Instruct across Neutral, Truthful, and Deceptive Prompts on **`sp_en_trans`**

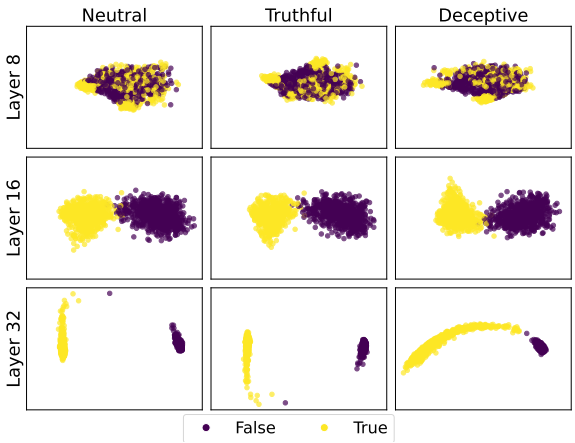

Figure 11: Layer-wise PCA visualization for Gemma-2-9B-Instruct on **`cities`**.